

ESG-Buddy: Agentic AI-Powered ESG Compliance Copilot

Abstract

ESG disclosure is now mandatory under BRSR (SEBI, India), GRI, TCFD, and SASB. Auditing a 400-page sustainability report against 150+ clauses takes consultants 40-80 hours per framework. No open tool offers clause-level, evidence-cited verdicts across all four frameworks. ESGBuddy closes this gap with a hybrid RAG + agentic LLM + deterministic rule pipeline.

Motivation

- BRSR mandatory for India's top 1,000 listed firms; scope expands annually.
- GRI, TCFD, SASB required simultaneously by institutional investors.
- RAG grounds LLM outputs in evidence, reducing hallucination (Lewis et al., 2020).
- Agentic self-reflection mirrors auditor reasoning with auditable chains of thought.

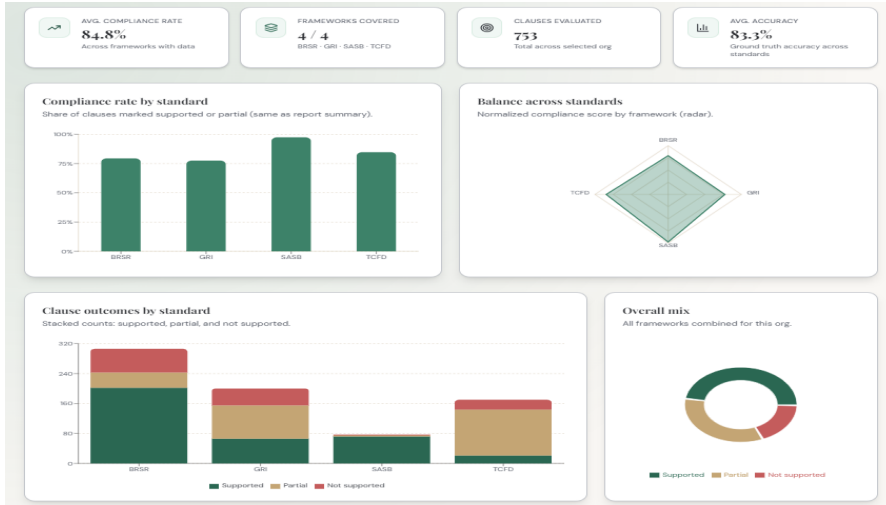


Fig. 01. Organisational dashboard(TCS): rate, radar, stacked outcomes, overall mix.

Literature Review & Objectives

RAG (Lewis et al., 2020) grounds generation in retrieved passages, reducing hallucination on regulatory text. Dense retrieval (Reimers & Gurevych, 2019) outperforms BM25 on domain-specific text. Chain-of-thought (Wei et al., 2022) and Reflexion (Shinn et al., 2023) enable iterative self-correction. Prior ESG tools target one framework at a time; clause-level, evidence-cited, multi-framework audit remains an open gap.

OBJECTIVES

- End-to-end clause-level ESG compliance across BRSR, GRI, SASB, TCFD.
- Hybrid pipeline: semantic retrieval, LLM reasoning, deterministic rule validation.
- Framework-specific prompting matching each standard's philosophy.
- Agentic chain-of-thought + self-reflection for audit-grade traceability.
- Benchmark on human-annotated ground truth (13 companies).
- Production-ready web UI with human-in-the-loop review.

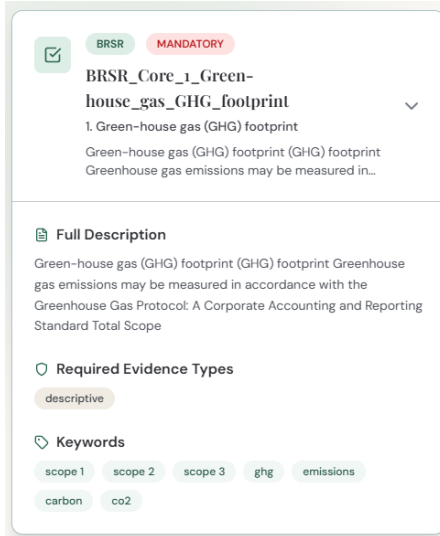


Fig. 02. ESG Clause Engine: description, evidence types, keywords.

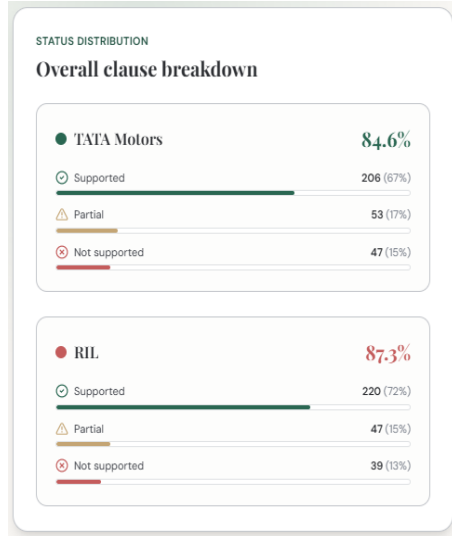


Fig. 03. Per-framework compliance cards (TATA Motors).

Methods & Implementation

STACK

FastAPI 0.109 backend (Python 3.11) · React 18 + Vite + Tailwind + Recharts frontend · GPT-4o-mini (temp 0.1) + text-embedding-3-small · ChromaDB 0.4.22 (company_documents, esg_clauses).

HYBRID PIPELINE (4 STAGES + AGENTIC LOOP)

- Semantic Retrieval — 512-token chunks, 50-token overlap, top-k dense, 0.12 min-similarity.
- LLM Reasoning — framework-specific CoT + Reflexion-style self-reflection.
- Rule Validation — deterministic numeric, temporal, keyword, existence checks.
- Classification — blended LLM + rule verdict (Supported / Partial / Not Supported) with calibrated confidence.

GROUND TRUTH & COMPANIES

13 companies × 4 frameworks = 52 benchmark pairs, hand-labelled clause-by-clause. Covers RIL, TCS, Infosys, TATA Motors, Sasken, Himadri, NYK, Nestlé, Givaudan, GPM, Unilever, Apple, Amazon.

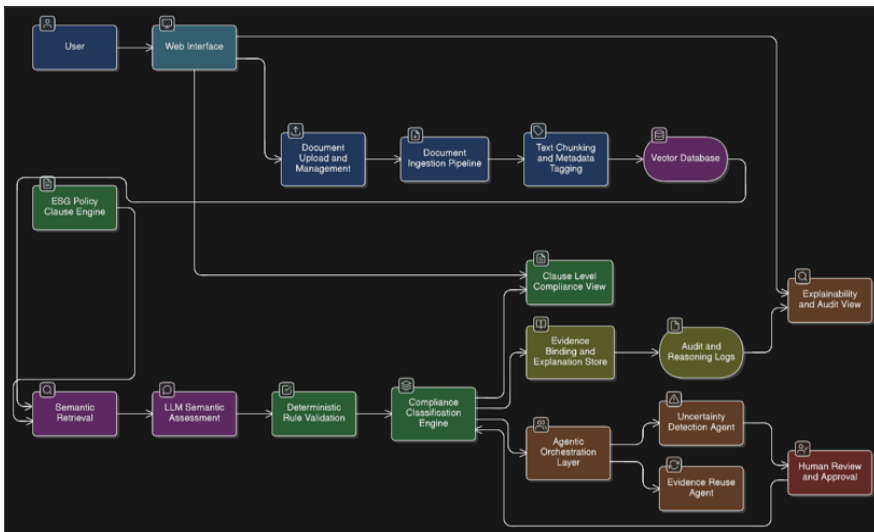


Fig. 04. End-to-end system architecture (React + FastAPI + ChromaDB + GPT-4o-mini).

Results & Discussion

HEADLINE METRICS

- 13 companies · 52 report-framework pairs
- Overall F1 **82.9%** · Status acc. **80.7%**
- Precision **83.6%** · Recall **82.2%**

FRAMEWORK-WISE F1

- BRSR **87.6** · SASB **84.4**
- GRI **81.9** · TCFD **77.6**

TCFD (77.6 F1) lags due to its qualitative, scenario-based disclosures forward-looking climate risk statements are harder to verify against evidence than structured numeric metrics.

Rule validation overrode LLM errors at >90% precision. 22–28% of low-confidence clauses routed to human review. Cost <\$0.40/report at ~2.1 s/clause.

BRSR's prescriptive, checkbox-style regulatory format (SEBI mandate) yields the highest F1 (87.6) structured disclosures are easier to match against retrieved evidence.

Conclusion & Future Scope

ESGBuddy delivers audit-grade, clause-level compliance verdicts across all four major ESG frameworks through a hybrid RAG + agentic LLM + deterministic rule pipeline, with transparent evidence and human-in-the-loop review.

LIMITATIONS & FUTURE WORK

- PDF extraction fidelity on multi-column and tabular layouts remains imperfect.
- Hosted LLM dependency motivates a fine-tuned ESG-domain model as next step.
- Active-learning loop planned to scale the human-labelled gold set.
- Cross-year delta tracking, compare a company's compliance across two annual reports to surface regression or improvement.

Key References

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